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The Role of Boundedness in Characterizing Shannon Entropy*

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This paper contains a characterization of Shannon's entropy measure $H(p_1, ..., p_n) = -\sum p_i \log p_i$ from the following properties: recursivity, 3-symmetry, normalization, and f is bounded on $[0, 1]$, where $f(x) = H(x, 1 - x)$ (boundedness). Two new techniques are presented: approximation of $\log N$ by rationals and iteration of the fundamental functional equation.

1. INTRODUCTION

Let Φ be the set of all finite discrete probability spaces, i.e., the elements of Φ are n-tuples $(p_1, ..., p_n)$, for $n = 1, 2, ...$, where $p_i \geqslant 0$ and $\sum_1^n p_i = 1$. According to Shannon (1948), the entropy (average self-information, average uncertainty, etc.) of an element $(p_1, ..., p_n)$ belonging to Φ is $H(p_1, ..., p_n) = -\sum p_i \log p_i$, where log is to the base 2. Our purpose is to characterize H in terms of the following axioms. Here we view H as a real-valued function acting on the elements $(p_1, ..., p_n)$ of Φ and, at the same time, for a fixed n with the p_i's $(1 \leqslant i \leqslant n)$ varying regard $H(p_1, ..., p_n)$ as a function of n variables. Ingarden and Urbanik (1962) take the same point of view of H in their characterization of information without probability; moreover, their Φ is a class of finite Boolean algebras.

AXIOM I (Recursivity). *Let* $(p_1, ..., p_n) \in \Phi$, $n \geqslant 2$, *and* $p_1 + p_2 > 0$. *Then*

$$H(p_1, ..., p_n) = H(p_1 + p_2, ..., p_n) + (p_1 + p_2)\, H(p_1/(p_1 + p_2), p_2/(p_1 + p_2)).$$

AXIOM II (3-Symmetry). *The function* $H(p_1, p_2, p_3)$ *is symmetric in* p_1, p_2, p_3, *where* $(p_1, p_2, p_3) \in \Phi$.

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AXIOM III (Normalization). *$f(\frac{1}{2}) = 1$, where $f(x) = H(x, 1 - x)$, $x \in [0, 1]$.*

AXIOM IV (Boundedness). *f is bounded on the unit interval I, i.e., there is a positive constant B such that $|f(x)| \leqslant B$ for all $x \in [0, 1]$.*

We prove, without recourse to the Erdös (1946) theorem on additive functions,

THEOREM 1. *$H(p_1, ..., p_n) = -\sum p_i \log p_i$ for $(p_1, ..., p_n) \in \Phi$ iff recursivity, 3-symmetry, normalization, and boundedness hold.*

This improves a result of Daróczy and Kátai (1970) who assume, beside Axioms I–IV, that f is nonnegative. Previous authors required stronger analytic properties on f than boundedness; Faddeev (1956), continuity; Tverberg (1958), Lebesgue integrability on $[0, 1]$; Lee (1964), Lebesgue measurability on $(0, 1)$; and Daróczy (1964), smallness for small probabilities. Also, see the survey paper, Aczél (1970), or for a comprehensive and up to date treatment see the forthcoming book, Aczél and Daróczy (1975).

The aim of characterizing entropy with as few properties as possible, aside from the interest attached to the functional equation aspects (see Aczél, 1966), stems from the problem of extracting a suitable concept of information, intrinsically and independently from probabilistic considerations. Currently many people (see for example Forte (1970) and Domotor (1970)) recognize that the concept of information itself is, perhaps, a more fundamental concept than probability. Theorem 1 suggests that, perhaps, algebraic properties (e.g. symmetry, recursivity) play a greater role than rather weak analytic properties (e.g. boundedness and nonnegativity) in such characterizations of information without probability. In other words, there is some promise in obtaining a characterization of information with no explicit probabilistic structure by relying more on algebraic properties than analytic properties. However, if Theorem 1 is regarded as a problem dealing with a certain type of functional equation, one would have to say that the algebraic properties and analytic properties carry equal weight.[1]

The proof of Theorem 1 is carried out in several stages. Only the reverse implication is presented since the other implication is clear. Section 2 is devoted to the determination of f on the rationals. The usual method is to apply the Erdös theorem on additive arithmetic functions. We give a new proof based on a well-known number theoretic result on approximating irrationals by rationals. In Section 3, another new approach is presented. The fundamental functional equation is iterated and we find that f becomes

[1] Since some regularity condition is necessary, see Lee (1964).

"approximately" locally Lipschitz near zero. Combining this fact with the "correct" form of f on the rationals, it is shown that f is small for small probabilities thus reducing the problem to Daróczy (1969). For the sake of completeness, Daróczy's proof is given in Section 4 completing the proof of the theorem. Section 5 deals with several concluding remarks.

Before we enter into the proof, we need to make some further comments about our notation. Throughout the paper, log is to the base 2, all n-tuples $(p_1, \ldots, p_n)$ belong to Φ, obvious summation ranges are suppressed, the letter B is reserved for the bound on f (Axiom IV), and again, $f(x) = H(x, 1-x)$ for $x \in [0, 1]$. Furthermore, most of our terminology is borrowed from Aczél and Daróczy (1975).

2. f On the Rationals

Lemma 2.1. *The functions $H(p_1, \ldots, p_n)$ are symmetric functions. Moreover, the function $f(x) = H(x, 1-x)$ for $x \in [0, 1]$ satisfies the fundamental functional equation and boundary conditions*

$$f(x) + (1-x) f\left(\frac{y}{1-x}\right) = f(y) + (1-y) f\left(\frac{x}{1-y}\right) \tag{1}$$

whenever $x, y \in [0, 1], \quad x + y \leqslant 1,$

$$f(1-x) = f(x)(x \in [0, 1]), \qquad H(1) = f(1) = 0, \qquad f(\tfrac{1}{2}) = 1.$$

Furthermore,

$$H(p_1, \ldots, p_n) = \sum (p_1 + \cdots + p_k) f\left(\frac{p_k}{p_1 + \cdots + p_k}\right), \tag{2}$$

if $p_1 + \cdots + p_k > 0$, *for* $k = 1, \ldots, n$. *Also,*

$$H(0, p_2, \ldots, p_n) = H(p_2, \ldots, p_n).$$

This lemma follows from recursivity, 3-symmetry, and normalization (see Borges (1967), Feinstein (1958), and Aczél and Daróczy (1975)).

Lemma 2.2 (Strong Additivity). *Let $\lambda = (p_1, \ldots, p_n)$ be an n-tuple in Φ. Split up λ into m ($m \geqslant 1$) disjoint subtuples λ_k ($1 \leqslant k \leqslant m$), so that in a natural sense we may write $\lambda = (\lambda_1, \ldots, \lambda_m)$. Also, for each $1 \leqslant k \leqslant m$ let P_k denote the sum of the p_i's in the sub-tuple λ_k. If $P_k > 0$ for $1 \leqslant k \leqslant m$, then*

$$H(\lambda) = H(P_1, \ldots, P_m) + \sum P_k H(\lambda_k / P_k).$$

Remark. In later application, we shall refer to the subtuples λ_k as "groups." Also, one should note that this lemma is merely a disguised form of strong additivity (see Aczél and Daróczy, 1975).

The proof of the lemma follows from recursivity and the symmetry of $H(p_1, \ldots, p_n)$ (see Ash, 1967); Aczél and Daróczy, 1975).

COROLLARY 2.1 (Additivity). *Let M, N denote positive integers and put $\phi(M) = H(1/M, \ldots, 1/M)$. Then*

$$\phi(MN) = \phi(M) + \phi(N),$$
$$\phi(N^k) = k\phi(N), \text{ and}$$
$$\phi(2^k) = k, \qquad \text{where } k \text{ is a nonnegative integer.}$$

The proof follows directly from the strong additivity property and the boundary condition of Eq. (1) in Lemma 2.1.

LEMMA 2.3. *There is a positive constant K such that*

$$|\phi(N)| \leqslant K \log N$$

for all integers $N \geqslant 1$.

Proof. Given a positive integer N, write N in binary form, so

$$N = 2^{m_1} + \cdots + 2^{m_s},$$

where $m_1 > m_2 > \cdots > m_s \geqslant 0$. Recall $H(1/N, \ldots, 1/N) = \phi(N)$. Form s groups: 2^{m_1} of $1/N$'s,..., 2^{m_s} of $1/N$'s, then by strong additivity we obtain

$$\phi(N) = H\left(\frac{2^{m_1}}{N}, \ldots, \frac{2^{m_s}}{N}\right) + \sum \frac{2^{m_i} m_i}{N}, \tag{3}$$

using $\phi(2^{m_i}) = m_i$. Since f is bounded by B, we obtain from (2) in Lemma 2.1

$$\left| H\left(\frac{2^{m_1}}{N}, \ldots, \frac{2^{m_s}}{N}\right)\right| \leqslant (s-1)B.$$

Clearly, $s - 1 \leqslant m_1 \leqslant \log N$ and $\sum(2^{m_i} m_i)/N \leqslant m_1$. Thus, taking absolute values in (3), the result is

$$|\phi(N)| \leqslant (s-1)\, B + m_1 \leqslant (\log N)\, B + \log N$$

or

$$|\phi(N)| \leqslant (B+1) \log N.$$

The lemma follows by taking $K = B + 1$.

Next we state a well-known result from number theory on approximating irrationals by rationals.

LEMMA 2.4. *Let θ be an irrational. There is an infinite number of rationals m/k (m, k are positive integers) such that*

$$\left| \theta - \frac{m}{k} \right| < \frac{1}{k^2}.$$

For proof see Hardy and Wright (1960), Chap. XI.

COROLLARY 2.2. *Let $N > 1$ be an odd integer. Either the exponential diophantine equation with a growth restriction on the remainder term,*

$$\begin{gathered} N^k = 2^m + c_m \quad \text{with} \\ 0 < c_m < 2^m(2^{1/k} - 1), \text{ where } c_m \text{ is an integer remainder term,} \end{gathered} \tag{4}$$

has an infinite number of solutions in positive integers m, k or, similarly, the equation

$$\begin{gathered} 2^m = N^k + c_m \quad \text{with} \\ 0 < c_m < N^k(2^{1/k} - 1) \end{gathered} \tag{5}$$

has an infinite number of solutions in positive integers m, k. Furthermore,

$$\begin{gathered} \lim \frac{m}{k} = \log N, \qquad \lim \frac{2^m}{N^k} = \lim \frac{N^k}{2^m} = 1 \\ \lim m = \infty \qquad \text{and} \qquad \lim k = \infty, \end{gathered} \tag{6}$$

where m, k run over an infinite set of solutions satisfying either (4) *or* (5).

Proof. Let $N \geqslant 3$ be a fixed odd integer. Then $\log N$ is an irrational. For, if $\log N = m/k$, then $N^k = 2^m$ which is clearly impossible. Thus, our number theoretic approximation lemma applies and we conclude that

$$|\log N - (m/k)| < 1/k^2 \tag{7}$$

has an infinite number of solutions. Hence, either $0 < \log N - (m/k) < 1/k^2$ or $0 < (m/k) - \log N < 1/k^2$ has infinitely many solutions. Suppose, for example, the latter possibility holds. Then, $0 < \log 2^m/N^k < 1/k$, hence

$$1 < 2^m/N^k < 2^{1/k}, \tag{8}$$

has infinitely many solutions in positive integers m, k.

Therefore, for each solution we can write $2^m = N^k + c_m$, where c_m is a positive integer remainder term. So, $2^m/N^k = 1 + (c_m/N^k)$ and substituting this in (8) results in

$$0 < c_m < N^k(2^{1/k} - 1),$$

which proves (5). The proof of (4) is similar; and the proof of (6) is clear from (7) and (8). This concludes the proof of the lemma.

LEMMA 2.5. *For every positive integer N we have*

$$\phi(N) = \log N.$$

Proof. Let $N > 1$ be a fixed odd integer. Apply the preceding corollary and suppose that (4) holds, i.e., $N^k = 2^m + c_m$ with $0 < c_m < 2^m(2^{1/k} - 1)$ has infinitely many solutions in positive integers m, k. For such m, k, form one group of 2^m, $1/N^k$'s and another group of c_m, $1/N^k$'s, then by strong additivity we obtain

$$H\left(\frac{1}{N^k}, \dots, \frac{1}{N^k}\right) = H\left(\frac{2^m}{N^k}, \frac{c_m}{N^k}\right) + \frac{2^m}{N^k} H\left(\frac{1}{2^m}, \dots, \frac{1}{2^m}\right) + \frac{c_m}{N^k} H\left(\frac{1}{c_m}, \dots, \frac{1}{c_m}\right).$$

Using the definition of f and ϕ, this becomes

$$\phi(N^k) = f(c_m/N^k) + (2^m/N^k)\,\phi(2^m) + (c_m/N^k)\,\phi(c_m).$$

But $\phi(2^m) = m$ and $\phi(N^k) = k\phi(N)$, and the above equation becomes

$$\phi(N) = \frac{f(c_m/N^k)}{k} + \frac{2^m}{N^k}\frac{m}{k} + \frac{c_m}{N^k}\frac{\phi(c_m)}{k}. \tag{9}$$

Clearly $[f(c_m/N^k)]/k \to 0$ (because f is bounded), $2^m/N^k \to 1$, and $m/k \to \log N$. Hence, it suffices to show that $(c_m/N^k)(\phi(c_m)/k) \to 0$ to obtain our result.

Lemma 2.3 implies that $|\phi(c_m)| \leqslant K \log c_m$, so

$$\left|\frac{c_m}{N^k}\frac{\phi(c_m)}{k}\right| \leqslant \frac{2^m}{N^k}(2^{1/k} - 1)K\frac{\log 2^m(2^{1/k} - 1)}{k}.$$

Simplifying the latter term gives

$$\frac{2^m}{N^k} K(2^{1/k} - 1)\left[\frac{m}{k} + \frac{1}{k}\log(2^{1/k} - 1)\right].$$

However, by l'Hospital's rule or by looking at series expansions, we find that $1/k \log (2^{1/k} - 1) \to 0$, as $k \to \infty$. Also $(2^{1/k} - 1) \to 0$. So indeed,

$$(c_m/N^k)(\phi(c_m)/k) \to 0.$$

Taking limits in (9) gives

$$\phi(N) = \log N.$$

If (5) in Corollary 2.4 holds, the proof is similar. If N is even, write $N = 2^s M$ where M is odd, then

$$\phi(N) = s + \phi(M) = \log 2^s + \log M = \log N.$$

This completes the proof.

COROLLARY 2.3. *We have*

$$f(q) = -q \log q - (1 - q) \log (1 - q) \tag{10}$$

for all rationals q in [0, 1]. *Furthermore,*

$$\lim_{q \to 0+} f(q) = 0, \tag{11}$$

for rationals $q \in (0, 1]$.

The proof of (10) is an easy application of Lemma 2.5 and strong additivity; clearly (11) is a direct consequence of (10).

3. ITERATION OF THE FUNDAMENTAL FUNCTIONAL EQUATION

Note. In the remainder of the paper, for $x \in [0, 1]$, write x' as an abbreviation for $1 - x$, i.e., $x' = 1 - x$.

LEMMA 3.1. *Let $N \geqslant 3$ be an integer and let $u_0, x_0, y_0, w_0 \in [0, 1]$. Assume that $u_0 = 1/N < x_0 < y_0 < 1/(N-1) = w_0$. Recursively define u_k, x_k, y_k, w_k, for $0 \leqslant k \leqslant N - 3$ as follows:*

$$\begin{aligned} u_{k+1} &= u_k/u_k', \\ x_{k+1} &= x_k/y_k', \\ y_{k+1} &= y_k/x_k', \\ w_{k+1} &= w_k/w_k'. \end{aligned} \tag{12}$$

Note: $w_k = u_{k+1}$, $u_0 = 1/N$, $w_0 = 1/(N-1)$. *Then,*

$$\begin{aligned} u_k &< x_k < y_k < w_k \leqslant \tfrac{1}{2}, \\ u_k' &> x_k' > y_k' > w_k', \quad \text{and} \\ x_k + y_k &< 1, \qquad \text{for} \quad 0 \leqslant k \leqslant N-3. \end{aligned} \tag{13}$$

Proof. The assertion is trivial for $k = 0$. Proceed by induction and assume that $0 \leqslant k < N - 3$ and $u_k < x_k < y_k < w_k$. Then

$$\begin{aligned} x_{k+1} - u_{k+1} &= \frac{x_k}{1-y_k} - \frac{u_k}{1-u_k} \\ &= \frac{(x_k - u_k) + u_k(y_k - x_k)}{(1-y_k)(1-u_k)} > 0, \end{aligned} \tag{14}$$

so $x_{k+1} > u_{k+1}$,

$$\begin{aligned} y_{k+1} - x_{k+1} &= \frac{y_k}{1-x_k} - \frac{x_k}{1-y_k} \\ &= \frac{(y_k - x_k)[1 - (x_k + y_k)]}{(1-x_k)(1-y_k)} > 0, \end{aligned} \tag{15}$$

thus $y_{k+1} > x_{k+1}$, and finally

$$\begin{aligned} w_{k+1} - y_{k+1} &= \frac{w_k}{1-w_k} - \frac{y_k}{1-x_k} \\ &= \frac{(w_k - y_k) + w_k(y_k - x_k)}{(1-w_k)(1-x_k)} > 0, \end{aligned} \tag{16}$$

whence $w_{k+1} > y_{k+1}$. This completes the inductive step and proves $u_k < x_k < y_k < w_k$ for $0 \leqslant k \leqslant N - 3$; the remaining inequalities of (13) are simple consequences of this. This completes the proof.

COROLLARY 3.1. *Under the same hypothesis as Lemma* 3.1 *define* $z_k = y_k - x_k > 0$, *for* $0 \leqslant k \leqslant N - 3$. *Then for* $0 \leqslant k \leqslant N - 3$,

$$y_0' y_1' \cdots y_k' \leqslant \frac{u_0}{u_{k+1}} = \frac{N-k-1}{N} \tag{17}$$

and for $1 \leqslant k \leqslant N - 3$,

$$y_0' \cdots y_{k-1}'(z_k/z_0) \leqslant (u_0')^k. \tag{18}$$

Proof. For $u_0 = 1/N$, it is easy to verify that $u_k = 1/(N - k)$. Also, from (12) we have $u_k' = u_k/u_{k+1}$. Inequality (13) implies that $y_0' \cdots y_k' \leqslant u_0' \cdots u_k'$. But $u_0' \cdots u_k' = (u_0/u_1)(u_1/u_2) \cdots (u_k/u_{k+1}) = u_0/u_{k+1}$ and $u_0/u_{k+1} = (N - (k + 1))/N$ which proves (17).

Now $y_0' \cdots y_{k-1}'(z_k/z_0) = (y_0' z_1/z_0) \cdots (y_{k-1}' z_k/z_{k-1})$ and (15) gives

$$y_{k-1}'\left(\frac{z_k}{z_{k-1}}\right) = \frac{1 - (x_{k-1} + y_{k-1})}{1 - x_{k-1}} = y_k'.$$

Hence $y_0' \cdots y_{k-1}'(z_k/z_0) = y_1' \cdots y_k' \leqslant (u_0')^k$ because (13) yields

$$u_0' > u_k' > y_k' \qquad \text{for} \quad 1 \leqslant k \leqslant N - 3$$

which proves (18). This ends the proof of the corollary.

LEMMA 3.2. *Let $N \geqslant 3$ be an integer. If $1/N < x_0 < y_0 < 1/(N - 1)$, then*

$$|f(x_0) - f(y_0)| \leqslant (4B/N) + |x_0 - y_0| NB, \tag{19}$$

where B is the bound on f.

Proof. The fundamental functional equation (see (1) in Lemma 2.1) is

$$f(x_0) + (1 - x_0) f(y_0/(1 - x_0)) = f(y_0) + (1 - y_0) f(x_0/(1 - y_0)).$$

Using the notation in Lemma 3.1 and Corollary 3.1, this may be written as

$$f(x_0) - f(y_0) = y_0'[f(x_1) - f(y_1)] - z_0 f(y_1). \tag{20}$$

More generally, after iterating $n = N - 3$ times, (20) becomes

$$\begin{aligned} f(x_0) - f(y_0) = y_0' \cdots y_n'&[f(x_{n+1}) - f(y_{n+1})] \\ &- z_0 f(y_1) - y_0' z_1 f(y_2) - \cdots \\ &- y_0' \cdots y_{k-1}' z_k f(y_{k+1}) - \cdots \\ &- y_0' \cdots y_{n-1}' z_n f(y_{n+1}). \end{aligned} \tag{21}$$

Taking absolute values in (21), using the bound on f, and taking into account (17) and (18) of Corollary 3.1, the result is

$$|f(x_0) - f(y_0)| \leqslant (4B/N) + z_0 B(1 + u_0' + (u_0')^2 + \cdots + (u_0')^n). \tag{22}$$

But

$$1 + (u_0') + \cdots + (u_0')^n < \sum_{k=0}^{\infty} (u_0')^k = 1/(1 - u_0') = 1/u_0 .$$

Since $u_0 = 1/N$, we obtain (19) from (22). This concludes the proof.

COROLLARY 3.2 (Small for Small Probabilities).

$$\lim_{x \to 0+} f(x) = 0. \tag{23}$$

Proof. Let $\epsilon > 0$ be given. By (11) in Corollary 2.3, we can choose N so large that for any rational y in $[0, 1/(N-1)]$ we have $|f(y)| < \epsilon/3$. Also, we can choose N large enough so that $4B/N < \epsilon/3$ and $1/N < \epsilon/3$. Let x be any irrational in $(0, 1/(N-1))$. Suppose that $1/M < x < 1/(M-1)$ where $M \geqslant N$. Next choose a rational y close enough to x so that $1/M < x < y < 1/(M-1)$ and $|x - y| < 1/BM^2$. Then

$$\begin{aligned} |f(x)| &= |f(x) - f(y) + f(y)| \\ &\leqslant |f(x) - f(y)| + |f(y)|. \end{aligned}$$

Applying Lemma 3.2 gives

$$|f(x)| \leqslant (4B/M) + (1/M) + |f(y)| < \epsilon/3 + \epsilon/3 + \epsilon/3 = \epsilon. \tag{24}$$

So $|f(x)| < \epsilon$ for any x in $[0, 1/(N-1)]$, and we are done.

4. DARÓCZY'S PROOF

The functional equation, again, is

$$f(x) + (1 - x)f(y/(1 - x)) = f(y) + (1 - y)f(x/(1 - y)), \tag{25}$$

where $x, y \in [0, 1]$, $x + y \leqslant 1$, and $f(x) = f(1 - x)$. The symmetry of f, $f(x) = f(1 - x)$, allows us to assume that $x \in [0, \frac{1}{2})$. Fix an irrational x in $(0, \frac{1}{2})$ and let q be a rational in $[0, \frac{1}{2}]$ such that $q \to x+$ (i.e., let q approach x from above). We can determine y with $y \to 0$ so that $q = x/(1 - y)$, $x + y \leqslant 1$, and $y > 0$. Therefore (25) can be written

$$f(x) - f(q) = f(y) - (1 - x)f(y/(1 - x)) - yf(q). \tag{26}$$

Clearly on the right side of (26), the first two terms go to zero by Corollary 3.2 and the last term goes to zero because f is bounded and $y \to 0$. Hence $\lim_{q\to x+} f(q) = f(x)$, thus Corollary 2.5 gives

$$f(x) = -x \log x - (1 - x) \log (1 - x).$$

Applying (2) in Lemma 2.1 finishes the proof of Theorem 1.

5. Concluding Remarks

Remark 1. In Lemma 2.4, it is worth noting that for each odd integer $N \geqslant 3$, $\log N$ is transcendental. This is almost a direct consequence of the Gelfond–Schneider theorem (see Hardy and Wright, 1960, p. 176) which states: for α, β algebraic, α different from 0 and 1, and β irrational, then α^β is transcendental. In our case, $\alpha = 2$, $\beta = \log N$, and $2^{\log N} = N$; thus, if $\log N$ is algebraic, then N is transcendental which is absurd. Hence $\log N$ is transcendental. Consequently one can expect that $\log N$ is approximated very well by rationals m/k, perhaps to an order $1/k^n$ for some $n > 2$ or possibly to an order with a more rapidly decreasing function of k.[2] Nonetheless, one observes from the proof of Lemma 2.5 that the order $1/k^2$ suffices for our purposes.

Remark 2. The proof of $\phi(N) = \log N$ (see Lemma 2.5) was motivated by the hope that for each odd integer $N \geqslant 3$ there are infinitely many positive integers m, k such that

$$N^k = 2^m + 1. \tag{27}$$

A very simple proof of Lemma 2.5 would result if this were true. Unfortunately, J. W. Mellender demonstrated that (27) has at most one solution m, k for each N except when $N = 3$ and then there exactly two solutions. Nevertheless, my point is this: A study of exponential diophantine equations (with perhaps growth restrictions on remainder terms) similar to (27) which have an infinite number of solutions, is closely related to problem of approximating $\log N$ by rationals and, in turn, with the problem of characterizing $\log N$.

Remark 3. Lemma 3.2, $|f(x_0) - f(y_0)| \leqslant (4B/N) + |x_0 - y_0| NB$ if $1/N < x_0 < y_0 < 1/(N-1)$ and $N \geqslant 3$, means that f is "approximately"

[2] *Note added in proof.* The reader will see in a future paper that a good *lower* bound estimate of $|\log N - (m/k)|$, as a quantitative measure of the transcendence or $\log N$, can also be exploited.

locally Lipschitz on $(0, \frac{1}{2})$, where the error term $4B/N \to 0$ for $N \to \infty$. Moreover, one observes that this is, in some sense, close to the actual

$$|f(x_0) - f(y_0)| \leqslant |x_0 - y_0| \left| \log \frac{x_0}{1 - x_0} \right|$$

because

$$\left| \log \frac{x_0}{1 - x_0} \right| < \left| \log \frac{1/N}{1 - (1/N)} \right| = \log(N - 1) < N.$$

Also from the proof of Lemma 3.2 (cf. Eq. (21)), there may be some hope in driving the error term $y_0' \cdots y_n'[f(x_{n+1}) - f(y_{n+1})]$ to zero by iterating infinitely often, provided one could find a way to "control" the sequences x_k, y_k. Then f would become locally Lipschitz. If this could be shown, the characterization proof could be completed without first determining f on the rationals. One applies the methods of Tverberg (1958), Kendall (1964), or Lee (1964). Also consult Aczél (1966) for general techniques in the solution of functional equations.

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